Liquid Extraction

Group #3

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Introduction

- Separating a contaminant (acetic acid) from white oil using water
- Two fluids are immiscible (don't mix)
- Water and white oil have different densities.

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Theory

- Mass Transfer
  - For a staged system, the concentration of the Solute in the feed for stage n+1 is the following:

$$y_{n+1} = \left(\frac{E^{n+1} - 1}{E - 1}\right) * \ y_1$$

y_1 = initial concentration of solute in feed, H = feed solvent, K = equilibrium constant

$E = \frac{KL}{H}$, L = extraction solvent

- Murphree Efficiency

$$E_{MR} = \left(\frac{X_S - X_{S+1}}{X_S - X^*_{S+1}}\right)$$

Where:
E_{MR} = Murphree efficiency of raffinate
X^*_{S+1} = composition of raffinate phase in equilibrium with extract phase leaving stage

$$E_{ME} = \left(\frac{Y_{S+1} - Y_S}{Y^*_{S+1} - Y_S}\right)$$

Where:
E_{ME} = Murphree efficiency of extract
Y^*_{S+1} = composition of extract phase in equilibrium with raffinate phase leaving stage

$$E_0 = \frac{N_T}{N_A}$$

Where:
E_0 = Overall efficiency
N_T = Theoretical number of stages required
N_A = Actual number of stages used

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Apparatus

- Three stages
- Each stage has a mixer, settler, and agitator
- Agitator can be set to different speeds
  - Use tachometer to measure the speed
- Fluid is moved to the next stage, and then product tank fills up.

Material & Supplies

- Buret and Buret Stand
- Stir bars and Stir plate
- Erlenmeyer flasks
- Large pipette (25 mL) / Bulb / Small pipette
- Parafilm
- Phenolphthalein Indicator
- NaOH 0.01M Solution for titration
- White Oil
- Acetic Oil
- House Water
- Colored tape / Sharpies to mark beakers

Procedure

- Take an initial sample of white oil to determine existing concentration of acetic acid
- Add enough acetic acid to create a 0.5wt% mixture of acetic acid and white oil
- Check to make sure the feed flow rate ratios are correct

- Choose a motor speed
- Wait 15 minutes to equilibrate
- For each stage, take a sample of water and sample of oil
- Cover samples with parafilm
- Titrate samples with NaOH
- Repeat above steps for 2 higher motor speeds

Experimental Challenges

- Samples must be the sample volume throughout the experiment
- Make sure your samples have indicator before titration
- Keep an eye on the level of white oil in the feed, you do not want to run out of oil.
- Keep the stages and feed tank are covered

Safety

- Wear goggles and gloves. Acetic acid can irritate skin
- Keep stages and feed tank covered. Do not inhale acetic acid
- Clean any spills on floor

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